

# An Nguyen, Dr.-Ing.

Digital Focus Topic Lead AI & Digital Assistants · Postdoctoral Researcher · AI Freelancer

Berlin, Germany · [an.nguyen@live.de](mailto:an.nguyen@live.de) · [website](#) · [LinkedIn](#) · [GitHub](#) · [Google Scholar](#)

## PROFESSIONAL SUMMARY

I bridge the gap between ML research and enterprise production — combining a Ph.D. in Computer Science (Applied Machine Learning in Healthcare and Industry), 10+ peer-reviewed publications, and 10+ years in AI/ML across research and industry (8+ years professional) with a track record of deploying AI at scale. At Siemens Healthineers I define AI strategy, architecture, governance, and enterprise adoption of agentic AI, LLMs, knowledge graphs, and multimodal AI for Customer Services globally. I turn research ideas into shipped products, build global AI communities and teams, and publish in top-tier venues — bringing both scientific rigor and pragmatic delivery to every role.

## CORE COMPETENCIES

**AI Strategy & Leadership:** AI Strategy · Enterprise AI Architecture · Technical Leadership · Product Ownership · AI Governance · Responsible AI · Stakeholder Management · Cross-functional Leadership · AI Roadmapping · Technology Strategy · Innovation Management

**Machine Learning & AI:** Agentic AI · Large Language Models (LLMs) · Retrieval-Augmented Generation (RAG) · Knowledge Graphs · Ontologies · Multimodal AI · NLP · Explainable AI · Probabilistic Machine Learning · Time Series Analysis · Predictive Analytics

**Engineering & MLOps:** Python · PyTorch · TensorFlow · Azure · Databricks · MLOps · SQL · Git · Linux · Cloud & On-Prem Infrastructure · ML System Design

**Product & Delivery:** Product Ownership · Agile Development · Technical Program Leadership · AI Solution Architecture · Production ML Systems · Enterprise Software · Digital Assistants · Customer Service AI

**Research & Healthcare:** Healthcare AI · Medical Devices · Clinical Data Analytics · IoT Analytics · Process Mining · Scientific Research · Experimental Design · Technology Transfer · Research Supervision · Mentoring

## PROFESSIONAL EXPERIENCE

### Digital Focus Topic Lead — AI & Digital Assistants

Jun 2024 – Present

#### Siemens Healthineers

- Lead AI strategy, architecture, and roadmap for Customer Services.
- Lead global cross-functional initiatives delivering agentic AI, LLM, multimodal AI, and digital-assistant solutions.
- Design reusable AI platforms using Azure, Databricks, knowledge graphs, semantic technologies, and on-prem infrastructure.
- Define governance for trustworthy AI, evaluation, explainability, and responsible AI.
- Partner with executives, architects, and engineering teams to translate AI research into enterprise products.

### Freelance AI Engineer

Mar 2024 – Present

#### Independent

- Build LLM, agentic-AI, and production-ML solutions for startups.

### Senior Data Scientist & Product Owner

Oct 2022 – May 2024

#### Siemens Healthineers

- Designed and deployed production ML systems processing near-real-time industrial machine data.
- Built operational-intelligence platforms with automated ticketing, monitoring, and predictive analytics.
- Co-organized an enterprise AI innovation challenge and hackathon.

### Data Scientist & Product Owner (part-time during Ph.D.)

Oct 2018 – Sep 2022

#### Siemens Healthineers

- Developed production ML solutions using IoT, enterprise, and PLM data.
- Built explainable-AI solutions for customer-sentiment prediction and operational analytics.
- Co-invented a U.S. patent for AI-driven customer-satisfaction prediction.

## ACADEMIC, RESEARCH & TEACHING

---

**Postdoctoral Researcher** — University of Erlangen–Nürnberg (FAU) 2023 – Present

Supervise PhD and Master's students; healthcare-AI research.

**Research Group Lead (commissarial) · Course Creator & Instructor** — FAU Erlangen–Nürnberg 2018 – 2022

Led Applied ML Workgroup (PhD students, industry projects, 2019–2020). Created and taught *ML & Data Analytics for Industry 4.0 Seminar* (2019–2022), *ML for Time Series Projects* (2020–2022), and *Project ML & Data Analytics* (2020–2022). Supervised 20+ Master's theses, projects, and internships.

**Doctoral Researcher** — University of Erlangen–Nürnberg (FAU) 2018 – 2023

Research in machine learning for healthcare and business analytics (magna cum laude).

**Visiting Researcher** — University of California, Irvine 2022

Probabilistic modelling of irregular and mixed-type time-series data.

**Visiting Researcher** — Frankel Cardiovascular Center & Biomedical & Clinical Informatics Lab, University of Michigan 2017

ML and signal-processing algorithms for atrial-fibrillation detection from ECG data.

**Peer review:** NeurIPS · IJCAI · ECML · IEEE BHI

## EDUCATION

---

**Dr.-Ing., Computer Science**

University of Erlangen–Nürnberg

Magna Cum Laude · 2018 – 2023

**M.Sc.Eng., Electrical & Computer Engineering**

University of Michigan

GPA 4.0 / 4.0 · Top 1% · 2016 – 2017

**Study Abroad — Electrical Engineering**

KTH Royal Institute of Technology, Stockholm

ERASMUS+ · 2014 – 2015

**M.Sc., Electrical Engineering**

Technical University of Berlin

Grade 1.0 · Top 1% · 2016 – 2018

**B.Sc., Electrical Engineering**

Technical University of Berlin

Grade 1.4 · Top 1% · 2011 – 2016

## SELECTED PUBLICATIONS (FULL LIST ON GOOGLE SCHOLAR)

---

- Schwinn et al. incl. **An Nguyen** (co-author). "Improving Robustness against Real-World and Worst-Case Distribution Shifts through Decision Region Quantification." *ICML 2022*.
- **An Nguyen** et al. (first author). "System Design for a Data-Driven and Explainable Customer Sentiment Monitor Using IoT and Enterprise Data." *IEEE Access*, 2021.
- **An Nguyen** et al. (first author). "Development and Clinical Validation of Inertial Sensor-Based Gait-Clustering Methods in Parkinson's Disease." *JNRE*, 2019.

## AWARDS, COMMUNITIES & SKILLS

---

- **Patent:** AI-based Customer Satisfaction Prediction for Medical Devices (US App. 17/375,306).
  - **Awards:** Outstanding Master's Degree Award — VDI Berlin-Brandenburg, best engineering graduates in the region (2019) · Fulbright Travel Grant — U.S. government merit award for graduate study at the University of Michigan (2016) · Rosa Luxemburg Foundation Scholarship — competitive full study scholarship by the German Federal Ministry of Education for gifted students (2011–2018).
  - **Communities:** Co-founded company internal Data Science Community (500+ members)
  - **Languages:** German (native) · English (fluent) · Swedish (proficient) · Vietnamese (proficient) · French (beginner).
  - **Engineering:** Python · PyTorch · TensorFlow · Azure · Databricks · LangChain · LangGraph · MLflow · SQL · Git · Linux.
- 

Professional references available upon request.